

Examiner
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2. Influenza, commonly known as “the flu”, is an infectious disease caused by the influenza virus, which infects the epithelial cells of the upper respiratory tract. The virus is composed of a protein capsid which surrounds the enzyme, RNA polymerase and its genetic material, RNA.
- (a) Describe the differences between the genetic material of the Influenza virus and the genetic material found within the nucleus of the epithelial cell it infects. [4]

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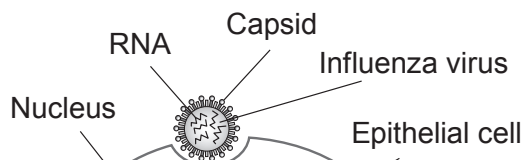
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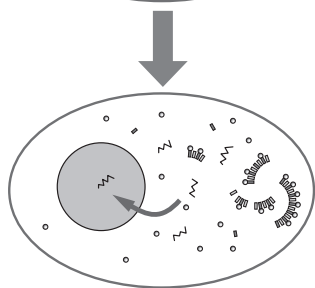
- (b) The virus can only replicate in living (host) cells, where it utilises nutrients and organelles within the cell to multiply quickly. The diagram below shows the stages in the replication of the virus.



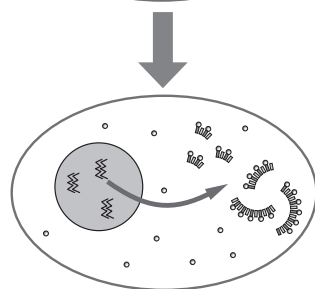
- 1 Influenza virus becomes attached to a target epithelial cell.



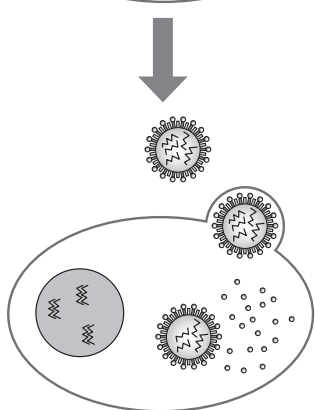
- 2 The cell engulfs the virus.



- 3 Viral contents are released. Viral RNA enters the nucleus where it is replicated by the viral RNA polymerase.



- 4 Viral mRNA is used to make viral proteins.



- 5 New viral particles are made and released into the extracellular fluid. The cell, which is not killed in the process, continues to make a new virus.



(i) State the name of the process occurring between stages 1 and 2. [1]

(ii) In stage 3, the capsid has broken down and the viral RNA and RNA polymerase enter the nucleus, where the viral RNA is replicated. Apply your knowledge of RNA polymerase to describe how the viral RNA is replicated. [4]

(iii) To complete the replication of the virus, capsids need to be produced (stage 4). Describe how the proteins in the capsid are produced. [5]



(iv) Explain why the virus is unable to synthesise its own capsid.

[2]

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